

A SELF-CONTAINED RESEARCH NOTE

Apéry Arrays and Conjectural Zeta Accelerations

Proofs for every shifted diagonal in OEIS A143007,
and both neighboring diagonals in A108625

September 20, 2026

Abstract

The OEIS array A143007 contains a conjectural one-parameter family of rapidly convergent series for $\zeta(3)$. Its main diagonal consists of the classical Apéry numbers. We prove the entire family, not merely a finite collection of numerical instances. The proof constructs a rational potential on the nonnegative integer lattice: two explicitly certified contiguous identities make its edge increments independent of the chosen path, and elementary estimates identify its limit as $\zeta(3)$. The same method proves both conjectural neighboring-diagonal formulas for $\zeta(2)$ in A108625 and extends them to every upper and lower offset. We also obtain exact finite identities, signed rational error bounds, all monotone-path expansions, the classical Apéry recurrences, and sharp fixed-offset asymptotics. The accompanying code checks 22,962 identities in exact arithmetic and independently verifies the finite symbolic certificates. The analytic proof does not depend on those tests.

Status of the claim. The targeted formulas are still described conjecturally in the OEIS versions consulted on the date above. This note supplies a complete proof of those statements. It does not claim that their underlying Apéry acceleration mechanism, or every consequence developed here, is new.

Contents

1	The problem and its mathematical setting	3
1.1	Exactly which conjectures are proved?	3
2	The arrays, their indexing, and the main results	3
3	Finite contiguous identities: the algebraic certificates	5
3.1	The $\zeta(3)$ square	5
3.2	The $\zeta(2)$ square	6
4	A rational potential proving the $\zeta(3)$ conjecture	7
4.1	Path independence on the integer lattice	7
4.2	Identifying the limit without assuming an Apéry theorem	8
4.3	The diagonal increment	8

5	A signed rational potential for the $\zeta(2)$ conjectures	9
6	Exact remainders and certified rational intervals	10
6.1	The positive $\zeta(3)$ error	10
6.2	The signed $\zeta(2)$ error	11
6.3	An elementary proof of geometric acceleration	11
6.4	Numerical certificates, not merely decimal agreement	12
7	Recovering the classical Apéry recurrences	12
7.1	Denominator recurrences	12
7.2	The companion numerator solutions	13
8	Sharp growth and error asymptotics	14
8.1	A discrete Laplace estimate	14
8.2	The T saddle	14
8.3	The C saddle and its asymmetry	15
8.4	Exact leading constants for the errors	16
8.5	Why a fixed row is slower than a diagonal	16
9	Reproducible verification and computational use	17
9.1	What was checked	17
9.2	How to reproduce the artifacts	18
9.3	A practical evaluation route	18
10	Conclusion and scope	18
A	Identification with the OEIS arrays and row generating functions	19
A.1	Two finite forms of the Legendre polynomial	19
A.2	The row generating function of C	19
A.3	A finite square identity and the row generating function of T	20
	References	21

1 The problem and its mathematical setting

Apéry's sequences are an especially appropriate setting for an OEIS conjecture: they connect explicit integer sums, second-order recurrences, rational approximation, and special values of the Riemann zeta function. The familiar sequence associated with $\zeta(3)$ is

$$A_n = \sum_{j=0}^n \binom{n}{j}^2 \binom{n+j}{j}^2 = 1, 5, 73, 1445, 33001, 819005, \dots, \quad (1.1)$$

where the displayed list begins at $n = 0$. This is [A005259](#). Its $\zeta(2)$ companion, [A005258](#), is

$$b_n = \sum_{j=0}^n \binom{n}{j}^2 \binom{n+j}{j} = 1, 3, 19, 147, 1251, 11253, \dots \quad (1.2)$$

These are not incidental integer sequences: they are the denominators in the classical Apéry approximation constructions [3, 4, 5, 6].

The two-variable arrays studied below have these sequences as their main diagonals. They also have an independent geometric interpretation. Let

$$\mathbf{A}_m = \{(x_1, \dots, x_{m+1}) \in \mathbb{Z}^{m+1} : \sum_i x_i = 0\}, \quad \|x\| = \frac{1}{2} \sum_i |x_i|.$$

The array A108625 counts balls in $\mathbf{A}_m$; A143007 counts balls in $\mathbf{A}_m \times \mathbf{A}_m$ for the sum of the two norms [2, 1, 7]. Our proof will use finite binomial sums rather than assume any geometric facts about these lattices.

1.1 Exactly which conjectures are proved?

The formula section of A143007 explicitly labels its shifted-diagonal $\zeta(3)$ formula a “Conjectural result for other diagonals” [1]. We prove it for every integer offset $k \geq 0$ in [theorem 2.1](#). The entry attributes its construction to Peter Bala and dates the original sequence to July 22, 2008.

A108625 describes two neighboring-diagonal $\zeta(2)$ formulas conjecturally in its comments of July 18, 2008 [2]. They are the $k = 1$ cases of [theorems 2.2](#) and [2.3](#). The theorems here also give every other nonnegative offset, with the necessary distinction between the upper and lower diagonals.

For reproducibility, the consulted internal revisions are A143007, revision 75, and A108625, revision 98; both carry a May 30, 2026 revision timestamp. A conjectural label in a database is evidence about the entry's wording, not a guarantee that no proof exists elsewhere. Two-dimensional and staircase versions of Apéry acceleration are classical; Cohen's account explicitly develops that viewpoint [5, §§3–4]. The contribution of this note is a direct, self-contained proof of the exact OEIS statements, together with explicit finite certificates and error analysis.

2 The arrays, their indexing, and the main results

All indices in this note are *square-array* indices. They are not indices in the one-dimensional antidiagonal flattening used on the OEIS pages. Set

$$L(m, j) = \binom{m}{j} \binom{m+j}{j}, \quad L(m, j) = 0 \quad \text{if } j > m, \quad (2.1)$$

and, for $m, n \geq 0$, define

$$T_{m,n} = \sum_{j=0}^{\min(m,n)} L(m, j)L(n, j), \quad (2.2)$$

$$C_{m,n} = \sum_{j=0}^{\min(m,n)} L(m, j) \binom{n}{j}. \quad (2.3)$$

The array T is A143007, and C is A108625, with the dimension parameter in the first index and the radius in the second. Indeed,

$$\binom{m+j}{2j} \binom{2j}{j} = L(m, j),$$

so (2.2) immediately agrees with the OEIS formula. The equivalence of (2.3) with the entry's binomial formula and row generating function is proved in [section A](#).

The initial portions are

$T_{m,n}$	$n = 0$	1	2	3	4	$C_{m,n}$	$n = 0$	1	2	3	4
$m = 0$	1	1	1	1	1	$m = 0$	1	1	1	1	1
1	1	5	13	25	41	1	1	3	5	7	9
2	1	13	73	253	661	2	1	7	19	37	61
3	1	25	253	1445	5741	3	1	13	55	147	309

Thus $T_{m,n} = T_{n,m}$, but generally $C_{m,n} \neq C_{n,m}$. In particular, transposing C silently would interchange the two $\zeta(2)$ conjectures and alter their constants and signs.

Both arrays are positive integers and are nondecreasing in each index, as follows directly from their nonnegative summands. Useful boundary values are

$$T_{0,n} = T_{m,0} = C_{0,n} = C_{m,0} = 1, \quad (2.4)$$

$$T_{m,1} = 1 + 2m(m+1), \quad C_{m,1} = 1 + m(m+1), \quad C_{1,n} = 1 + 2n. \quad (2.5)$$

Their diagonals are precisely $T_{n,n} = A_n$ and $C_{n,n} = b_n$.

Write

$$H_k^{(s)} = \sum_{j=1}^k \frac{1}{j^s}, \quad E_k = 2 \sum_{j=1}^k \frac{(-1)^{j+1}}{j^2}, \quad H_0^{(s)} = E_0 = 0.$$

Theorem 2.1 (All shifted $\zeta(3)$ diagonals). *For every integer $k \geq 0$,*

$$\boxed{\zeta(3) = H_k^{(3)} + \sum_{n=1}^{\infty} \frac{(2n+k)(3n^2+3nk+k^2)}{n^3(n+k)^3 T_{n-1,n+k-1} T_{n,n+k}}.} \quad (2.6)$$

The series is positive and converges geometrically. Its partial sum through $n = N$ is strictly below $\zeta(3)$.

At $k = 0$, (2.6) is the classical Apéry series

$$\zeta(3) = 6 \sum_{n=1}^{\infty} \frac{1}{n^3 A_{n-1} A_n}. \quad (2.7)$$

For example, the first genuinely shifted case is

$$\zeta(3) = 1 + \frac{21}{8 \cdot 13} + \frac{95}{216 \cdot 13 \cdot 253} + \cdots$$

Theorem 2.2 (All upper $\zeta(2)$ diagonals). *For every integer $k \geq 0$,*

$$\zeta(2) = H_k^{(2)} + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(n+k)^2 + (2n+k)^2}{n^2(n+k)^2 C_{n-1,n+k-1} C_{n,n+k}}. \quad (2.8)$$

The series is absolutely convergent. If $S_{N,k}^{(2,+)}$ denotes its partial sum including $H_k^{(2)}$, then $\text{sgn}(\zeta(2) - S_{N,k}^{(2,+)}) = (-1)^N$.

Theorem 2.3 (All lower $\zeta(2)$ diagonals). *For every integer $k \geq 0$,*

$$\zeta(2) = E_k + \sum_{n=1}^{\infty} (-1)^{n+k+1} \frac{n^2 + (2n+k)^2}{n^2(n+k)^2 C_{n+k-1,n-1} C_{n+k,n}}. \quad (2.9)$$

The series is absolutely convergent. If $S_{N,k}^{(2,-)}$ denotes its partial sum including E_k , then $\text{sgn}(\zeta(2) - S_{N,k}^{(2,-)}) = (-1)^{N+k}$.

At $k = 0$ both theorems give

$$\zeta(2) = 5 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2 b_{n-1} b_n}. \quad (2.10)$$

Their $k = 1$ cases are exactly the conjectures recorded in A108625:

$$\zeta(2) = 1 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(n+1)^2 + (2n+1)^2}{n^2(n+1)^2 C_{n-1,n} C_{n,n+1}}, \quad (2.11)$$

$$\zeta(2) = 2 - \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2 + (2n+1)^2}{n^2(n+1)^2 C_{n,n-1} C_{n+1,n}}. \quad (2.12)$$

For comparison with the entry, these begin

$$1 + \frac{13}{4 \cdot 5} - \frac{34}{36 \cdot 5 \cdot 37} + \cdots, \quad 2 - \frac{10}{4 \cdot 7} + \frac{29}{36 \cdot 7 \cdot 55} - \cdots.$$

3 Finite contiguous identities: the algebraic certificates

The proof is driven by identities involving the four corners of one unit square. There is no experimental recurrence guessing in what follows: each identity is certified by an explicit telescoping summand.

3.1 The $\zeta(3)$ square

Fix integers $m, n \geq 1$ and abbreviate

$$a = T_{m-1,n-1}, \quad b = T_{m,n-1}, \quad c = T_{m-1,n}, \quad d = T_{m,n}, \quad s = (m+n)(m^2 + mn + n^2).$$

Lemma 3.1. *The four corners satisfy*

$$m^3 d + n^3 a = s c, \quad (3.1)$$

$$n^3 d + m^3 a = s b. \quad (3.2)$$

Proof. Let $M = \min(m, n)$ and $f_j = L(m, j)L(n, j)$ for $0 \leq j \leq M$. The ratios needed to lower the parameters are

$$\frac{L(m-1, j)}{L(m, j)} = \frac{m-j}{m+j}, \quad \frac{L(n-1, j)}{L(n, j)} = \frac{n-j}{n+j}.$$

They remain valid at an endpoint where a numerator vanishes. Define

$$R_j = \frac{j^4}{(m+j)(n+j)} f_j \quad (0 \leq j \leq M), \quad R_{M+1} = 0.$$

The ratio of consecutive summands gives, for $0 \leq j \leq M$,

$$R_{j+1} = (m-j)(n-j)f_j. \quad (3.3)$$

For $j = M$ both sides vanish; for $j < M$ this follows from

$$\frac{f_{j+1}}{f_j} = \frac{(m-j)(m+j+1)(n-j)(n+j+1)}{(j+1)^4}.$$

The following rational identity is the certificate:

$$\begin{aligned} m^3 + n^3 \frac{(m-j)(n-j)}{(m+j)(n+j)} - s \frac{m-j}{m+j} \\ = 2(m+n) \left\{ \frac{j^4}{(m+j)(n+j)} - (m-j)(n-j) \right\}. \end{aligned} \quad (3.4)$$

It is checked simply by multiplying by $(m+j)(n+j)$ and expanding. Multiplication by f_j turns its right side into $2(m+n)(R_j - R_{j+1})$. Summing from 0 to M proves (3.1), since $R_0 = R_{M+1} = 0$. Interchanging m and n and using the symmetry of T proves (3.2). $\square$

3.2 The $\zeta(2)$ square

Now use the same local letters for the corners of C :

$$a = C_{m-1, n-1}, \quad b = C_{m, n-1}, \quad c = C_{m-1, n}, \quad d = C_{m, n}, \quad q = n^2 + (m+n)^2.$$

Lemma 3.2. *The four corners satisfy*

$$2n^2d - m^2a = qb, \quad (3.5)$$

$$m^2d + 2n^2a = qc. \quad (3.6)$$

Proof. Put $g_j = L(m, j)\binom{n}{j}$, $M = \min(m, n)$, and set

$$r_m = \frac{m-j}{m+j}, \quad r_n = \frac{n-j}{n}, \quad R_j = \frac{j^3}{m+j} g_j \quad (0 \leq j \leq M), \quad R_{M+1} = 0.$$

Here r_m and r_n are the ratios obtained by lowering m and n , respectively. As in the preceding proof,

$$R_{j+1} = (m-j)(n-j)g_j,$$

because

$$\frac{g_{j+1}}{g_j} = \frac{(m-j)(m+j+1)(n-j)}{(j+1)^3} \quad (j < M).$$

Two elementary rational identities are

$$m^2 + 2n^2 r_m r_n - q r_m = 2 \left\{ \frac{j^3}{m+j} - (m-j)(n-j) \right\}, \quad (3.7)$$

$$2n^2 - m^2 r_m r_n - q r_n = \frac{2(m+n)}{n} \left\{ \frac{j^3}{m+j} - (m-j)(n-j) \right\}. \quad (3.8)$$

Multiply each identity by g_j , sum, and use $R_0 = R_{M+1} = 0$. The resulting identities are (3.6) and (3.5). All denominators used here are nonzero because $m, n \geq 1$. $\square$

These three certificates are sufficient for the whole infinite family of conjectures. Their role is structural: a local identity will allow an infinite series to be evaluated by changing its lattice path.

4 A rational potential proving the $\zeta(3)$ conjecture

4.1 Path independence on the integer lattice

A horizontal edge into (m, n) is the step $(m-1, n) \rightarrow (m, n)$, and a vertical edge into (m, n) is $(m, n-1) \rightarrow (m, n)$. If their weights are $h(m, n)$ and $v(m, n)$, respectively, the square condition is

$$h(m, n-1) + v(m, n) = v(m-1, n) + h(m, n). \quad (4.1)$$

It says that the two routes from $(m-1, n-1)$ to (m, n) have the same weight.

Lemma 4.1 (Finite path independence). *If (4.1) holds for every $m, n \geq 1$, then every monotone lattice path from $(0, 0)$ to (m, n) has the same sum of edge weights. Consequently there is a potential $F_{m,n}$ with $F_{0,0} = 0$ and*

$$F_{m,n} - F_{m-1,n} = h(m, n), \quad F_{m,n} - F_{m,n-1} = v(m, n).$$

Proof. Encode a monotone path as a word with m horizontal steps and n vertical steps. Adjacent horizontal-vertical pairs can be interchanged without changing the sum, by (4.1). Such interchanges connect any two words with the same numbers of each step. Define $F_{m,n}$ as this common sum. Appending a last edge proves the asserted increments. $\square$

For T , assign the positive rational edge weights

$$h_3(m, n) = \frac{1}{m^3 T_{m-1,n} T_{m,n}}, \quad v_3(m, n) = \frac{1}{n^3 T_{m,n-1} T_{m,n}}. \quad (4.2)$$

Each expression is used only when the corresponding edge exists. With the local corner notation of theorem 3.1, condition (4.1), after clearing positive denominators, is

$$c(n^3 d + m^3 a) = b(m^3 d + n^3 a).$$

Both sides are $sb c$, by theorem 3.1. Hence the weights are path independent.

Let their potential be $U_{m,n}$. Taking the two boundary routes yields the explicit finite identity

$$U_{m,n} = H_n^{(3)} + \sum_{i=1}^m \frac{1}{i^3 T_{i-1,n} T_{i,n}} = H_m^{(3)} + \sum_{j=1}^n \frac{1}{j^3 T_{m,j-1} T_{m,j}}. \quad (4.3)$$

In particular $U_{m,n}$ is rational, symmetric, and strictly increasing when either coordinate increases by one.

4.2 Identifying the limit without assuming an Apéry theorem

Lemma 4.2. *For all $m, n \geq 0$, $U_{m,n} < \zeta(3)$. Moreover, $U_{m,n} \rightarrow \zeta(3)$ whenever $\max(m, n) \rightarrow \infty$.*

Proof. The second expression in (4.3) and monotonicity of T imply

$$0 \leq U_{m,n} - H_m^{(3)} \leq \frac{\zeta(3)}{T_{m,1}} = \frac{\zeta(3)}{1 + 2m(m+1)}.$$

This tends to zero as $m \rightarrow \infty$, uniformly in n ; thus for each fixed n , the strictly increasing sequence $U_{m,n}$ tends to $\zeta(3)$. In particular every finite value is strictly below $\zeta(3)$. The two expressions in (4.3) give

$$H_{\max(m,n)}^{(3)} \leq U_{m,n} < \zeta(3).$$

The last assertion now follows by squeezing. $\square$

It follows, incidentally, that *every* infinite monotone path starting at $(0, 0)$ yields a positive series with sum $\zeta(3)$. Rows and diagonals are particular paths, not separate approximation phenomena.

4.3 The diagonal increment

The decisive cancellation is finite. A horizontal step followed by a vertical step, together with (3.2), gives

$$\begin{aligned} U_{m,n} - U_{m-1,n-1} &= \frac{1}{m^3 ab} + \frac{1}{n^3 bd} \\ &= \frac{n^3 d + m^3 a}{m^3 n^3 abd} \\ &= \frac{(m+n)(m^2 + mn + n^2)}{m^3 n^3 T_{m-1,n-1} T_{m,n}}. \end{aligned} \quad (4.4)$$

The intermediate corner b cancels because the contiguous identity makes its numerator a multiple of b . This explains the apparently special cubic polynomial in the conjectured summand.

Proof of theorem 2.1. Apply (4.4) at the points $(n, n+k)$ and sum the resulting identity for $n = 1, \dots, N$. Since $U_{0,k} = H_k^{(3)}$, the exact result is

$$U_{N,N+k} = H_k^{(3)} + \sum_{n=1}^N \frac{(2n+k)(3n^2 + 3nk + k^2)}{n^3(n+k)^3 T_{n-1,n+k-1} T_{n,n+k}}. \quad (4.5)$$

Let $N \rightarrow \infty$ and apply theorem 4.2. All terms are positive, and the partial sums are below the limit. Geometric convergence follows from the explicit bounds in section 6, independently of the sharper asymptotics in section 8. $\square$

5 A signed rational potential for the $\zeta(2)$ conjectures

The array C is not symmetric, and the correct weights necessarily remember the first coordinate:

$$h_2(m, n) = \frac{2(-1)^{m+1}}{m^2 C_{m-1,n} C_{m,n}}, \quad v_2(m, n) = \frac{(-1)^m}{n^2 C_{m,n-1} C_{m,n}}. \quad (5.1)$$

The square condition (4.1), after clearing denominators and a common sign, becomes

$$c(2n^2d - m^2a) = b(m^2d + 2n^2a).$$

By [theorem 3.2](#), both sides are qbc . Hence these weights define a rational potential $V_{m,n}$. The boundary values now differ on the two axes:

$$V_{0,n} = H_n^{(2)}, \quad V_{m,0} = E_m.$$

The two boundary routes give

$$V_{m,n} = H_n^{(2)} + 2 \sum_{i=1}^m \frac{(-1)^{i+1}}{i^2 C_{i-1,n} C_{i,n}} = E_m + (-1)^m \sum_{j=1}^n \frac{1}{j^2 C_{m,j-1} C_{m,j}}. \quad (5.2)$$

Lemma 5.1. *The potential $V_{m,n}$ tends to $\zeta(2)$ whenever $\max(m, n) \rightarrow \infty$.*

Proof. First, $E_m \rightarrow \zeta(2)$: separating the even terms in the absolutely convergent series gives

$$\sum_{j \geq 1} \frac{(-1)^{j+1}}{j^2} = \zeta(2) - 2 \sum_{j \geq 1} \frac{1}{(2j)^2} = \frac{1}{2} \zeta(2).$$

The second expression in (5.2) gives the uniform estimate

$$|V_{m,n} - E_m| \leq \frac{\zeta(2)}{C_{m,1}} = \frac{\zeta(2)}{1 + m(m+1)}. \quad (5.3)$$

On the other hand, the first expression gives

$$|V_{m,n} - H_n^{(2)}| \leq \frac{2\zeta(2)}{C_{1,n}} = \frac{2\zeta(2)}{2n+1}. \quad (5.4)$$

For (5.4), use $C_{i,n} \geq C_{1,n}$ and $C_{i-1,n} \geq 1$ for every $i \geq 1$. Either estimate proves the limit when its corresponding coordinate tends to infinity, uniformly in the other. Together they prove the assertion for $\max(m, n) \rightarrow \infty$. $\square$

As before, every infinite monotone path produces a convergent expansion for the same constant, although the edge weights now have signs. To combine the two edges of a square, use (3.5):

$$\begin{aligned} V_{m,n} - V_{m-1,n-1} &= \frac{2(-1)^{m+1}}{m^2 ab} + \frac{(-1)^m}{n^2 bd} \\ &= (-1)^{m+1} \frac{2n^2 d - m^2 a}{m^2 n^2 a b d} \\ &= (-1)^{m+1} \frac{n^2 + (m+n)^2}{m^2 n^2 C_{m-1,n-1} C_{m,n}}. \end{aligned} \quad (5.5)$$

Proof of theorems 2.2 and 2.3. Sum (5.5) along the points $(n, n + k)$, starting from $(0, k)$. This gives the exact finite identity

$$V_{N, N+k} = H_k^{(2)} + \sum_{n=1}^N (-1)^{n+1} \frac{(n+k)^2 + (2n+k)^2}{n^2(n+k)^2 C_{n-1, n+k-1} C_{n, n+k}}. \quad (5.6)$$

Instead sum along $(n+k, n)$, starting from $(k, 0)$. The result is

$$V_{N+k, N} = E_k + \sum_{n=1}^N (-1)^{n+k+1} \frac{n^2 + (2n+k)^2}{n^2(n+k)^2 C_{n+k-1, n-1} C_{n+k, n}}. \quad (5.7)$$

Take $N \rightarrow \infty$ and apply theorem 5.1. Absolute convergence already follows by bounding the positive arrays below by 1: for fixed k , the absolute summands are $O(n^{-2})$. The exact signs of the errors follow from (6.4) below. $\square$

The additive constant E_k in the lower formula is not interchangeable with $H_k^{(2)}$. Likewise its extra sign $(-1)^k$ is not optional. Both are forced by the boundary values and the orientation of the potential.

6 Exact remainders and certified rational intervals

A proof of convergence is stronger when it also gives computable error bounds. Here the potentials make the error an explicit tail in either coordinate direction, so no numerical evaluation of a zeta function is required for certification.

6.1 The positive $\zeta(3)$ error

For fixed n , extend the horizontal path from (m, n) to infinity. By theorem 4.2,

$$\zeta(3) - U_{m, n} = \sum_{i=m+1}^{\infty} \frac{1}{i^3 T_{i-1, n} T_{i, n}} = \sum_{j=n+1}^{\infty} \frac{1}{j^3 T_{m, j-1} T_{m, j}}. \quad (6.1)$$

Both representations are positive. For $m \geq 1$, monotonicity gives

$$\begin{aligned} \frac{1}{(m+1)^3 T_{m, n} T_{m+1, n}} &\leq \zeta(3) - U_{m, n} \\ &\leq \frac{\zeta(3) - H_m^{(3)}}{T_{m, n} T_{m+1, n}} < \frac{1}{2m^2 T_{m, n} T_{m+1, n}}. \end{aligned} \quad (6.2)$$

The last inequality uses $\sum_{i>m} i^{-3} < \int_m^{\infty} t^{-3} dt = 1/(2m^2)$. There is a second bound with the coordinates interchanged. Thus, for $m, n \geq 1$, the explicitly rational number

$$B_3(m, n) = \min \left\{ \frac{1}{2m^2 T_{m, n} T_{m+1, n}}, \frac{1}{2n^2 T_{m, n} T_{m, n+1}} \right\} \quad (6.3)$$

satisfies

$$U_{m, n} < \zeta(3) < U_{m, n} + B_3(m, n).$$

For the conjectural series, substitute $(m, n) = (N, N + k)$.

6.2 The signed $\zeta(2)$ error

Taking a vertical tail in (5.2) gives the exact identity

$$\zeta(2) - V_{m,n} = (-1)^m \sum_{j=n+1}^{\infty} \frac{1}{j^2 C_{m,j-1} C_{m,j}}. \quad (6.4)$$

Every summand on the right after removal of $(-1)^m$ is strictly positive. Hence the sign is exactly $(-1)^m$, not merely eventually so. For $n \geq 1$, the vertical estimates are

$$\frac{1}{(n+1)^2 C_{m,n} C_{m,n+1}} \leq |\zeta(2) - V_{m,n}| < \frac{1}{n C_{m,n} C_{m,n+1}}. \quad (6.5)$$

The upper bound follows from $\sum_{j>n} j^{-2} < 1/n$.

There is a complementary horizontal tail:

$$\zeta(2) - V_{m,n} = 2 \sum_{i=m+1}^{\infty} \frac{(-1)^{i+1}}{i^2 C_{i-1,n} C_{i,n}}. \quad (6.6)$$

Its magnitudes strictly decrease to zero, since i^2 strictly increases and both array factors are nondecreasing. The alternating-series bound therefore gives

$$|\zeta(2) - V_{m,n}| \leq \frac{2}{(m+1)^2 C_{m,n} C_{m+1,n}}. \quad (6.7)$$

For $n \geq 1$, define

$$B_2(m, n) = \min \left\{ \frac{2}{(m+1)^2 C_{m,n} C_{m+1,n}}, \frac{1}{n C_{m,n} C_{m,n+1}} \right\}. \quad (6.8)$$

Then $V_{m,n}$ and $V_{m,n} + (-1)^m B_2(m, n)$ are endpoints of a closed rational interval containing $\zeta(2)$. The parity gives the correct ordering of its endpoints. The same construction works when $n = 0$ by using only the first term in (6.8).

6.3 An elementary proof of geometric acceleration

The bounds above already imply geometric convergence, without any saddle-point analysis. Since the central binomial coefficient is the largest of the $2N + 1$ coefficients whose sum is 4^N ,

$$\binom{2N}{N} \geq \frac{4^N}{2N+1}.$$

The $j = N$ summand in $T_{N,N+k}$ gives

$$T_{N,N+k} \geq \binom{2N}{N}^2 \geq \frac{16^N}{(2N+1)^2}.$$

Apply (6.2) and monotonicity to obtain, for $N \geq 1$,

$$0 < \zeta(3) - U_{N,N+k} < \frac{(2N+1)^4}{2N^2 256^N}. \quad (6.9)$$

Likewise, both $C_{N,N+k}$ and $C_{N+k,N}$ are at least $\binom{2N}{N}$. Using (6.7) gives

$$\max\{|\zeta(2) - V_{N,N+k}|, |\zeta(2) - V_{N+k,N}|\} \leq \frac{2(2N+1)^2}{(N+1)^2 16^N}. \quad (6.10)$$

These deliberately simple bounds are uniform over all offsets $k \geq 0$. The sharp exponential rates for fixed k are substantially faster.

6.4 Numerical certificates, not merely decimal agreement

table 1 reports rational bounds (6.3) and (6.8) computed by exact integer arithmetic. The displayed scientific notation has been rounded *upward*; the exact fractions are included in `data/certified_intervals.csv`. The full CSV also records the rational partial sums, the error signs, and decimal displays.

Table 1: Rigorous absolute-error upper bounds after N diagonal terms. Each displayed value is rounded upward to three significant digits.

Family	k	$N = 10$	$N = 20$	$N = 40$
$\zeta(3)$	0	$5.27 \cdot 10^{-30}$	$2.26 \cdot 10^{-60}$	$2.44 \cdot 10^{-121}$
	1	$1.79 \cdot 10^{-31}$	$7.15 \cdot 10^{-62}$	$7.45 \cdot 10^{-123}$
	3	$3.99 \cdot 10^{-34}$	$1.05 \cdot 10^{-64}$	$8.48 \cdot 10^{-126}$
$\zeta(2)$, upper	0	$4.57 \cdot 10^{-21}$	$5.88 \cdot 10^{-42}$	$9.46 \cdot 10^{-84}$
	1	$7.18 \cdot 10^{-22}$	$8.91 \cdot 10^{-43}$	$1.41 \cdot 10^{-84}$
	3	$2.69 \cdot 10^{-23}$	$2.59 \cdot 10^{-44}$	$3.53 \cdot 10^{-86}$
$\zeta(2)$, lower	0	$4.57 \cdot 10^{-21}$	$5.88 \cdot 10^{-42}$	$9.46 \cdot 10^{-84}$
	1	$2.80 \cdot 10^{-22}$	$3.44 \cdot 10^{-43}$	$5.40 \cdot 10^{-85}$
	3	$1.64 \cdot 10^{-24}$	$1.50 \cdot 10^{-45}$	$2.00 \cdot 10^{-87}$

For instance, the unshifted $\zeta(3)$ series after 40 terms has absolute error less than $2.44 \cdot 10^{-121}$. This is an absolute-error statement; claiming that a certain decimal string is correctly rounded would additionally require testing the certified interval against the relevant rounding boundary. The archive preserves exact fractions so that such tests can be made without relying on a floating-point zeta implementation.

7 Recovering the classical Apéry recurrences

The square identities also explain why these arrays contain the familiar Apéry denominators. This provides an independent consistency check and connects the proof with the usual one-dimensional formulation.

7.1 Denominator recurrences

On the diagonal of theorem 3.1,

$$T_{n,n-1} = T_{n-1,n} = \frac{A_n + A_{n-1}}{6} \quad (n \geq 1). \quad (7.1)$$

At the square with upper-right corner $(n+1, n)$, (3.1) gives

$$(n+1)^3 T_{n+1,n} + n^3 T_{n,n-1} = (2n+1)(3n^2 + 3n + 1)A_n.$$

Substitute (7.1). After multiplication by 6 and collection of terms, the result is

$$(n+1)^3 A_{n+1} = (34n^3 + 51n^2 + 27n + 5)A_n - n^3 A_{n-1}, \quad n \geq 1. \quad (7.2)$$

The initial values are $A_0 = 1$ and $A_1 = 5$.

Similarly, the two diagonal identities for C give

$$C_{n,n-1} = \frac{2b_n - b_{n-1}}{5}, \quad C_{n-1,n} = \frac{b_n + 2b_{n-1}}{5}. \quad (7.3)$$

Use (3.6) at $(m, n) = (n+1, n)$:

$$(n+1)^2 C_{n+1,n} + 2n^2 C_{n,n-1} = (n^2 + (2n+1)^2)b_n.$$

Substitution from (7.3) gives

$$(n+1)^2 b_{n+1} = (11n^2 + 11n + 3)b_n + n^2 b_{n-1}, \quad n \geq 1, \quad (7.4)$$

with $b_0 = 1$ and $b_1 = 3$. These are the classical recurrences recorded for A005259 and A005258 [3, 4].

7.2 The companion numerator solutions

Define rational arrays

$$P_{m,n} = T_{m,n} U_{m,n}, \quad R_{m,n} = C_{m,n} V_{m,n}. \quad (7.5)$$

They satisfy exactly the same local linear identities as T and C , respectively. For example, abbreviate the four U values by U_a, U_b, U_c, U_d . Then

$$\begin{aligned} m^3 d U_d + n^3 a U_a - s c U_c &= m^3 d (U_d - U_c) + n^3 a (U_a - U_c) \\ &= m^3 d h_3(m, n) - n^3 a v_3(m-1, n) \\ &= \frac{1}{c} - \frac{1}{c} = 0. \end{aligned}$$

The other identity follows by symmetry. For R , the corresponding cancellations are

$$\begin{aligned} 2n^2 d V_d - m^2 a V_a - q b V_b &= 2n^2 d v_2(m, n) + m^2 a h_2(m, n-1) = 0, \\ m^2 d V_d + 2n^2 a V_a - q c V_c &= m^2 d h_2(m, n) - 2n^2 a v_2(m-1, n) = 0. \end{aligned}$$

Thus the same elimination used for the denominators applies to their numerator companions.

Set $p_n = P_{n,n}$ and $r_n = R_{n,n}$. Then p_n satisfies (7.2), with $p_0 = 0, p_1 = 6$, and r_n satisfies (7.4), with $r_0 = 0, r_1 = 5$. In particular,

$$\frac{p_n}{A_n} = U_{n,n} \longrightarrow \zeta(3), \quad \frac{r_n}{b_n} = V_{n,n} \longrightarrow \zeta(2).$$

The adjacent determinant identities (often called Casoratians) are immediate from the diagonal increments:

$$p_n A_{n-1} - p_{n-1} A_n = \frac{6}{n^3}, \quad (7.6)$$

$$r_n b_{n-1} - r_{n-1} b_n = \frac{5(-1)^{n+1}}{n^2}. \quad (7.7)$$

The normalization here allows rational numerator sequences; no claim that all p_n or r_n are integers is intended. In the standard irrationality arguments, additional denominator-control statements are important. They are not needed to prove any of the series identities in this note.

8 Sharp growth and error asymptotics

The previous sections already prove the conjectures and give rigorous rational error bounds. We now determine the exact fixed-offset asymptotics, both to quantify the acceleration and to distinguish the effect of an offset from the effect of taking more diagonal terms.

Put

$$\tau = 1 + \sqrt{2}, \quad q_3 = \tau^2, \quad \lambda_3 = \tau^4 = 17 + 12\sqrt{2}, \quad (8.1)$$

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \lambda_2 = \varphi^5 = \frac{11 + 5\sqrt{5}}{2}, \quad (8.2)$$

$$c_3 = \frac{q_3}{2^{9/4}\pi^{3/2}}, \quad c_2 = \frac{\varphi^{5/2}}{2\pi 5^{1/4}}. \quad (8.3)$$

Theorem 8.1 (Fixed-offset array growth). *For every fixed integer $k \geq 0$, as $N \rightarrow \infty$,*

$$T_{N,N+k} \sim c_3 q_3^k \lambda_3^N N^{-3/2}, \quad (8.4)$$

$$C_{N,N+k} \sim c_2 \varphi^{2k} \lambda_2^N N^{-1}, \quad (8.5)$$

$$C_{N+k,N} \sim c_2 \varphi^{3k} \lambda_2^N N^{-1}. \quad (8.6)$$

8.1 A discrete Laplace estimate

We use the following elementary form of the saddle-point principle. Suppose f has a unique interior maximum at $t_0 \in (0, 1)$, $f''(t_0) < 0$, and the summands near $j/N = t_0$ have the uniform form

$$N^{-a} g(j/N) e^{Nf(j/N)} (1 + o(1)), \quad g(t_0) > 0.$$

If summands outside every fixed neighborhood of t_0 are bounded by a polynomial in N times $e^{N(f(t_0)-\delta)}$ for some $\delta > 0$, then their sum is asymptotic to

$$g(t_0) \sqrt{\frac{2\pi}{|f''(t_0)|}} N^{1/2-a} e^{Nf(t_0)}. \quad (8.7)$$

To see this, expand $f(t) = f(t_0) + \frac{1}{2}f''(t_0)(t - t_0)^2 + O(|t - t_0|^3)$ and use $x = (j - Nt_0)/\sqrt{N}$. The local sum becomes a Riemann sum for the Gaussian integral with mesh $N^{-1/2}$; the complement is negligible by the strict maximum and a Gaussian majorant. First restrict to a fixed bounded x interval, then let that interval expand. This also justifies multiplying the summands by a fixed continuous function of j/N : the leading constant is multiplied by its value at t_0 .

8.2 The T saddle

For $j/N = t$ in a compact subinterval of $(0, 1)$, Stirling's formula gives

$$L(N, j) \sim \frac{1}{2\pi N t} \sqrt{\frac{1+t}{1-t}} e^{N\ell(t)}, \quad (8.8)$$

where

$$\ell(t) = (1+t) \log(1+t) - (1-t) \log(1-t) - 2t \log t.$$

The exponent for $T_{N,N}$ is $f_3(t) = 2\ell(t)$, and its amplitude is $N^{-2}g_3(t)$ with

$$g_3(t) = \frac{1+t}{4\pi^2 t^2(1-t)}.$$

Since

$$\ell'(t) = \log \frac{1-t^2}{t^2},$$

the unique maximum is at $t_3 = 1/\sqrt{2}$. Direct substitution gives

$$e^{f_3(t_3)} = \lambda_3, \quad f_3''(t_3) = -8\sqrt{2}, \quad g_3(t_3) = \frac{q_3}{2\pi^2}.$$

Equation (8.7) now gives $c_3 \lambda_3^N N^{-3/2}$.

For a fixed offset, the exact ratio is

$$\frac{L(N+k, j)}{L(N, j)} = \prod_{r=1}^k \frac{N+j+r}{N-j+r}. \quad (8.9)$$

Near t_3 , this tends uniformly to $((1+t)/(1-t))^k$, whose value at t_3 is q_3^k . This proves (8.4).

For completeness, the endpoint estimates required by (8.7) follow directly from factorial bounds for the binomial summands. Their logarithms equal $N f_3(j/N) + O(\log N)$ uniformly when the entropy terms are extended continuously to 0 and 1. For fixed k , the ratio (8.9) is bounded by a polynomial in N on the whole summation range. Since f_3 has a strict interior maximum, the endpoints and every region separated from t_3 are exponentially smaller than the saddle contribution.

8.3 The C saddle and its asymmetry

Multiplying (8.8) by Stirling's approximation for $\binom{N}{j}$ gives exponent

$$f_2(t) = (1+t) \log(1+t) - 2(1-t) \log(1-t) - 3t \log t$$

and amplitude $N^{-3/2}g_2(t)$, where

$$g_2(t) = \frac{\sqrt{1+t}}{(2\pi)^{3/2} t^{3/2} (1-t)}.$$

The stationary equation is

$$f_2'(t) = \log \frac{(1+t)(1-t)^2}{t^3} = 0, \quad 1-t-t^2 = 0.$$

Thus the unique maximum occurs at $t_2 = 1/\varphi$. At that point,

$$e^{f_2(t_2)} = \lambda_2, \quad f_2''(t_2) = -\sqrt{5} \varphi^3,$$

and (8.7) gives $c_2 \lambda_2^N N^{-1}$.

For the upper diagonal, the offset multiplies the summand by

$$\frac{\binom{N+k}{j}}{\binom{N}{j}} = \prod_{r=1}^k \frac{N+r}{N-j+r} \rightarrow (1-t)^{-k},$$

which equals φ^{2k} at t_2 . For the lower diagonal the multiplier is (8.9), with limiting value $((1+t_2)/(1-t_2))^k = \varphi^{3k}$. The same endpoint argument as for T applies. This proves [theorem 8.1](#), including the asymmetric constants in its last two formulas.

8.4 Exact leading constants for the errors

Let $S_{N,k}^{(3)} = U_{N,N+k}$, and retain the notation $S_{N,k}^{(2,+)} = V_{N,N+k}$ and $S_{N,k}^{(2,-)} = V_{N+k,N}$.

Theorem 8.2 (Sharp fixed-offset errors). *For each fixed $k \geq 0$,*

$$\zeta(3) - S_{N,k}^{(3)} \sim \frac{96\sqrt{2}\pi^3}{\lambda_3^k(\lambda_3^2 - 1)} \lambda_3^{-2N}, \quad (8.10)$$

$$\zeta(2) - S_{N,k}^{(2,+)} \sim (-1)^N \frac{20\pi^2\sqrt{5}}{\varphi^{4k}(\lambda_2^2 + 1)} \lambda_2^{-2N}, \quad (8.11)$$

$$\zeta(2) - S_{N,k}^{(2,-)} \sim (-1)^{N+k} \frac{20\pi^2\sqrt{5}}{\varphi^{6k}(\lambda_2^2 + 1)} \lambda_2^{-2N}. \quad (8.12)$$

In the signed formulas, $\sim$ means that the ratio to the displayed signed expression tends to 1.

Proof. Let $a_{n,k}$ be the positive summand in (2.6). By [theorem 8.1](#),

$$a_{n,k} \sim \frac{6\lambda_3}{c_3^2 q_3^{2k}} \lambda_3^{-2n} = \frac{96\sqrt{2}\pi^3}{\lambda_3^k} \lambda_3^{-2n}.$$

The powers n^{-3} in the rational prefactor and in the product of array asymptotics cancel. Summing the resulting geometric tail gives (8.10).

For the upper $\zeta(2)$ series, the absolute value of the n th summand is asymptotic to

$$\frac{5\lambda_2}{c_2^2 \varphi^{4k}} \lambda_2^{-2n} = \frac{20\pi^2\sqrt{5}}{\varphi^{4k}} \lambda_2^{-2n}.$$

For the lower series, replace φ^{4k} by φ^{6k} and include the additional sign $(-1)^k$. An alternating geometric tail has denominator $1 + \lambda_2^{-2}$, which yields (8.11) and (8.12).

To justify these tail operations directly, if $a_n/(K\rho^n) \rightarrow 1$ with $0 < \rho < 1$, then $\sup_{n \geq N} |a_n/(K\rho^n) - 1| \rightarrow 0$. The absolute error incurred by replacing the entire tail with the corresponding geometric tail is therefore $o(\rho^N)$, in both the positive and alternating cases. No exchange of an uncontrolled asymptotic expansion with a sum is used. $\square$

The asymptotic decimal gain per extra diagonal term is

$$2 \log_{10} \lambda_3 = 3.0622054827 \dots \quad \text{for } \zeta(3), \quad 2 \log_{10} \lambda_2 = 2.0898764025 \dots \quad \text{for } \zeta(2).$$

A fixed offset changes the leading constant, not these exponential rates. Increasing k also increases the size and cost of the array entries, so the prefactor improvement alone does not establish an optimal computational strategy.

8.5 Why a fixed row is slower than a diagonal

For comparison, fix m and let $n \rightarrow \infty$. The leading binomial terms give

$$T_{m,n} \sim \alpha_m n^{2m}, \quad \alpha_m = \frac{\binom{2m}{m}}{(m!)^2}; \quad C_{m,n} \sim \beta_m n^m, \quad \beta_m = \frac{\binom{2m}{m}}{m!}.$$

Using the exact vertical tails and the elementary asymptotic $\sum_{j>n} j^{-s} \sim n^{1-s}/(s-1)$ for $s > 1$, we obtain

$$\zeta(3) - U_{m,n} \sim \frac{n^{-(4m+2)}}{\alpha_m^2(4m+2)}, \quad (8.13)$$

$$\zeta(2) - V_{m,n} \sim (-1)^m \frac{n^{-(2m+1)}}{\beta_m^2(2m+1)}. \quad (8.14)$$

Thus a fixed row gives increasing algebraic acceleration as the row index grows, whereas a fixed-offset diagonal gives geometric acceleration. Both behaviors are consequences of the same potential.

9 Reproducible verification and computational use

The proof is finite algebra followed by explicit convergence estimates. The accompanying computations are a separate audit of indexing, normalization, and signs. In particular, the proof of an infinite identity is not inferred from decimal agreement.

9.1 What was checked

The program `verify.py` uses only Python's standard library, with arbitrary-precision integers and rational fractions. Its default run checks the following exact equalities:

Test group	Equalities
Square identities, closed edges, and diagonal increments	12,800
Potentials, symmetry, and independent array formulas	8,405
Finite diagonal formulas, offsets $0 \leq k \leq 12$	1,599
Denominator and numerator recurrences	156
Published diagonal prefixes	2
Total	22,962

The square tests use $1 \leq m, n \leq 40$. The potential and independent array tests use $0 \leq m, n \leq 40$. Finite diagonal formulas are checked for every $0 \leq N \leq 40$ and $0 \leq k \leq 12$, including all three orientations/families and the empty-sum cases.

For additional independence, C is compared with the coefficient formula proved in [section A](#), and T is compared with the alternative OEIS sum

$$\sum_{j=0}^{\min(m,n)} \binom{m}{j}^2 \binom{m+n-j}{m}^2$$

[1]. These comparisons are finite implementation tests; the proof of the conjectures uses (2.2) and (2.3) directly.

The optional program `symbolic_certificates.py` uses SymPy to reduce the three rational certificates (3.4), (3.7), (3.8), and the two recurrence-elimination coefficients, to zero. All five tests pass. The supplied record identifies the tested SymPy version. These are symbolic identities, not evaluations at sample numerical points.

9.2 How to reproduce the artifacts

From the archive directory, run

```
python verify.py --max-index 40 --offsets 12 --output-dir data
python make_error_table.py
python symbolic_certificates.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The symbolic command requires SymPy; it is not required for the first two commands or for compiling the supplied article. The optional `numerical_checks.py` compares the exact partial sums with high-precision zeta values and illustrates the asymptotic constants. Its output is not used as an error certificate.

The CSV interval data store numerators and denominators rather than only decimal approximations. For $\zeta(3)$, use the endpoints S and $S + B$. For $\zeta(2)$, the recorded sign $\sigma \in \{1, -1\}$ gives endpoints S and $S + \sigma B$, ordered increasingly. The article, source notes, test programs, and generated data are all included in the archive.

9.3 A practical evaluation route

A direct array entry is a finite sum of $\min(m, n) + 1$ positive integers. The summand ratios in [section 3](#) allow successive summands to be generated without repeatedly evaluating factorials. For a fixed diagonal and N terms, the direct implementation uses $O(N^2)$ summand-generation steps when k is fixed; this counts arithmetic operations, not bit complexity. The integers grow exponentially in value and hence linearly in bit length along a fixed-offset diagonal.

For the main diagonals, [\(7.2\)](#) and [\(7.4\)](#) reduce denominator generation to $O(N)$ recurrence steps. The same recurrences for p_n and r_n produce the associated rational approximants. Whichever method is used, one can stop when the rational bound is below a prescribed absolute tolerance. The positive binomial formulas are particularly convenient for independent verification and avoid relying on cancellation in a recurrence.

10 Conclusion and scope

The conjectural family in A143007 is true for every nonnegative offset. Both neighboring-diagonal conjectures in A108625 are also true, and belong to two full offset families. The proof does not assume any of the conjectural formulas, a known diagonal limit, or Apéry's irrationality theorem. Instead it identifies the limit from the ordinary convergent series for $\zeta(3)$ and $\zeta(2)$ on the coordinate axes.

The structural reason is a closed rational system of edge increments. A finite telescoping certificate yields each square identity; the square identities yield path independence; a two-edge diagonal step cancels its intermediate array entry; and the boundary limits identify the sum. This chain of implications explains the cubic and quadratic numerators in the OEIS formulas, as well as the differing constants and signs in the two $\zeta(2)$ orientations.

The conclusions concern the stated acceleration formulas; no new irrationality theorem or resolution of unrelated Apéry-polynomial conjectures is claimed.

A Identification with the OEIS arrays and row generating functions

This appendix verifies the array identification algebraically and explains the Legendre-polynomial structure behind the crystal-ball interpretation. It is independent of the potential proof.

A.1 Two finite forms of the Legendre polynomial

Define the Legendre polynomial by Rodrigues' formula,

$$P_m(z) = \frac{1}{2^m m!} \frac{d^m}{dz^m} (z^2 - 1)^m.$$

Expansion at $z = 1$ gives

$$P_m(1 + 2u) = \sum_{j=0}^m L(m, j) u^j. \quad (\text{A.1})$$

Indeed, after substituting $z = 1 + 2u$, Rodrigues' formula becomes $m!^{-1} (d/du)^m [u^m (1+u)^m]$, whose coefficient of u^j is $\binom{m}{j} \binom{m+j}{j}$.

Applying the Leibniz rule instead gives

$$P_m(z) = 2^{-m} \sum_{i=0}^m \binom{m}{i}^2 (z-1)^i (z+1)^{m-i}. \quad (\text{A.2})$$

With $W_m(x) = \sum_{i=0}^m \binom{m}{i}^2 x^i$, this implies

$$P_m\left(\frac{1+x}{1-x}\right) = \frac{W_m(x)}{(1-x)^{m+1}}. \quad (\text{A.3})$$

A.2 The row generating function of C

Using $\sum_{n \geq 0} \binom{n}{j} x^n = x^j / (1-x)^{j+1}$ in (2.3), we obtain, as a formal power series,

$$\begin{aligned} \sum_{n \geq 0} C_{m,n} x^n &= \frac{1}{1-x} \sum_{j=0}^m L(m, j) \left(\frac{x}{1-x}\right)^j \\ &= \frac{1}{1-x} P_m\left(\frac{1+x}{1-x}\right) = \frac{W_m(x)}{(1-x)^{m+1}}. \end{aligned} \quad (\text{A.4})$$

Extracting the coefficient of x^n gives

$$C_{m,n} = \sum_{i=0}^{\min(m,n)} \binom{m}{i}^2 \binom{m+n-i}{m}. \quad (\text{A.5})$$

This is exactly the defining finite sum and row generating function listed in A108625 [2]. In particular, the orientation of the two indices has been verified, rather than inferred from a short sequence prefix.

A.3 A finite square identity and the row generating function of T

We next prove the finite identity

$$P_m(z)^2 = \sum_{j=0}^m L(m, j) \binom{2j}{j} \left(\frac{z^2 - 1}{4} \right)^j. \quad (\text{A.6})$$

It is a terminating Legendre square identity. A short differential verification is included to keep the identification self-contained.

Put $M = m(m + 1)$ and $Q(z) = P_m(z)^2$. The Legendre equation $(1 - z^2)P_m'' - 2zP_m' + MP_m = 0$, obtained by differentiating Rodrigues' formula, yields the following equation by the product rule:

$$\begin{aligned} (z^2 - 1)^2 Q''' + 6z(z^2 - 1)Q'' \\ + ((6 - 4M)z^2 + 4M - 2)Q' - 4MzQ = 0. \end{aligned} \quad (\text{A.7})$$

Since $P_m(-z) = (-1)^m P_m(z)$, Q is a polynomial in z^2 . Set $t = (z^2 - 1)/4$ and write $Q(z) = y(t)$. Substitution in (A.7) and cancellation as a polynomial identity gives

$$t^2(4t + 1)y''' + (18t^2 + 3t)y'' + ((12 - 4M)t + 1)y' - 2My = 0. \quad (\text{A.8})$$

For $y(t) = \sum_{j \geq 0} q_j t^j$, coefficient comparison gives

$$(j + 1)^3 q_{j+1} = 2(2j + 1)(m - j)(m + j + 1)q_j, \quad q_0 = P_m(1)^2 = 1.$$

The coefficients $q_j = L(m, j) \binom{2j}{j}$ satisfy exactly this recurrence and initial value, and vanish for $j > m$. This proves (A.6).

Now

$$\sum_{n \geq 0} L(n, j) x^n = \binom{2j}{j} \frac{x^j}{(1 - x)^{2j+1}},$$

so (2.2) and (A.6) imply

$$\begin{aligned} \sum_{n \geq 0} T_{m,n} x^n &= \frac{1}{1 - x} \sum_{j=0}^m L(m, j) \binom{2j}{j} \left(\frac{x}{(1 - x)^2} \right)^j \\ &= \frac{1}{1 - x} P_m \left(\frac{1 + x}{1 - x} \right)^2 = \frac{W_m(x)^2}{(1 - x)^{2m+1}}. \end{aligned} \quad (\text{A.9})$$

These are the row generating functions recorded in A143007. The passage from $W_m(x)/(1 - x)^m$ to its square is exactly what the sum norm on a product lattice requires for sphere counts; division by $1 - x$ then converts sphere counts to ball counts [1, 7].

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A143007, square array for the crystal-ball sequences of $A_n \times A_n$, originally contributed by Peter Bala, July 22, 2008. Internal revision 75, May 30, 2026. Consulted September 20, 2026. <https://oeis.org/A143007/internal>.
- [2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A108625, square array for the crystal-ball sequences of A_n ; relevant zeta-acceleration comments by Peter Bala, July 18, 2008. Internal revision 98, May 30, 2026. Consulted September 20, 2026. <https://oeis.org/A108625/internal>.
- [3] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A005259, Apéry numbers $\sum_{j=0}^n \binom{n}{j}^2 \binom{n+j}{j}^2$. Consulted September 20, 2026. <https://oeis.org/A005259>.
- [4] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A005258, Apéry numbers $\sum_{j=0}^n \binom{n}{j}^2 \binom{n+j}{j}$. Consulted September 20, 2026. <https://oeis.org/A005258>.
- [5] Henri Cohen, *Apéry Acceleration of Continued Fractions*, arXiv:2401.17720, 2024. In particular, Sections 3–4 discuss staircase/diagonal acceleration and the examples $\zeta(2)$ and $\zeta(3)$. <https://arxiv.org/abs/2401.17720>.
- [6] Armin Straub, *Multivariate Apéry numbers and supercongruences of rational functions*, *Algebra & Number Theory* **8** (2014), no. 8, 1985–2008. doi:10.2140/ant.2014.8.1985. Preprint: <https://arxiv.org/abs/1401.0854>.
- [7] Roland Bacher, Pierre de la Harpe, and Boris Venkov, *Séries de croissance et polynômes d'Ehrhart associés aux réseaux de racines*, *Annales de l'Institut Fourier* **49** (1999), no. 3, 727–762. doi:10.5802/aif.1689. https://www.numdam.org/item/AIF_1999__49_3_727_0/.